成都理工大學 中英合作办学
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OXFORD BROOKES UNIVERSITY

UNDERGRADUATE PROJECT PROPOSAL

Project Title	ProsePal:An Intelligent Writing Assistant for Web Novel Authors
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30/10/2025

Date Submitted

Table of Contents

Chapter 1. Introduction

1.1 Background of Study

With the rise of digital media, web literature has become a vibrant and significant component of the global cultural industry, attracting hundreds of millions of readers and creators [1]. However, despite this creative boom, many authors still rely on general-purpose office software such as Microsoft Word, Google Docs for their work. While powerful, these tools were designed for general document processing and fall short of meeting the complex demands of long-form narrative works, especially web novels, which require functionalities like non-linear chapter management, visualization of character relationships, and maintenance of plot consistency [2].

Creators often face numerous challenges, including frequently switching between multiple applications to consult research materials, manage ideas, and organize outlines, which significantly fragments their creative focus and reduces efficiency. Furthermore, the manuscript revision process often relies on subjective communication between authors and editors, lacking data-driven analytical tools to support decision-making, leading to inefficient revisions. The advancement of modern computational linguistics and human-computer interaction offers a potential solution to these pain points [3]. Therefore, developing a writing assistant system specifically designed for web novel authors—integrating intelligent analysis and efficient collaboration—is of significant practical importance and applicational value.

1.2 Project Aim

The primary aim of this project is to design and implement an integrated software solution called "ProsePal." This system is intended to significantly enhance the creative efficiency, work quality, and collaborative experience of web novel authors by providing a one-stop platform for ideation, writing, analysis, and collaboration.

1.3 Project Objectives

To achieve the project aim, the following types of work will be performed:

- Research and Analysis Work: To conduct a comparative analysis of existing writing tools to define functional baselines and identify key market gaps.
- System Design Work: To perform system and software architecture design,

including the creation of database schemas, API specifications, and high-fidelity UI mockups.

- Implementation and Development Work: To program the front-end and back-end modules of the application, including the core text editor, document management system, and intelligent analysis etc.
- Integration Work: To integrate the core application with third-party services and libraries, specifically for Natural Language Processing functionalities and real-time collaboration features.
- Testing and Quality Assurance Work: To execute a comprehensive testing strategy, including the development of unit tests, the execution of integration tests, and the coordination of User Acceptance Testing to ensure software robustness and reliability.

1.4 Project Overview

This project aims to develop an intelligent writing assistant software specifically designed for web novel creators. It combines traditional text editing functionalities with advanced data analysis, project management, and online collaboration tools to address the efficiency bottlenecks and creative obstacles that authors currently face in their workflow.

1.4.1 Scope

The software will provide a focused, integrated environment through the following detailed functionalities:

Intelligent Document Management:

- Real-time Cloud Synchronization: All work will be auto-saved and synced across devices.
- Version History: The system will automatically save snapshots of the document, allowing users to review, compare , and restore previous versions.
- Visual Chapter/Outline Management: A dedicated panel will allow authors to view their manuscript as a collection of "cards", which can be reordered via drag-and-drop to restructure the plot.

Integrated Research Environment:

- Split-Screen View: An embedded browser will allow writers to conduct research within the same window as their manuscript, eliminating context-switching.

- Research/Note Manager: A "corkboard" or notes section will allow users to clip text, images, and links from the web, and link them to specific chapters or characters.

NLP-Powered Intelligent Analysis:

- Character Relationship Mapping: The system will automatically parse the text to identify character interactions and generate a visual graph of their relationships.
- Sentiment Analysis: A "Story Emotion Curve" will be generated, plotting the emotional valence of the narrative chapter by chapter to help authors manage pacing.
- Consistency Checker: Authors can create "character sheets" with key traits.The tool will flag potential contradictions found later in the text.

Seamless Collaboration:

- Role-Based Access Control: Project owners can invite collaborators as "Editors" (full edit/comment rights) or "Viewers" (read-only).
- Suggestion Mode: Similar to "Track Changes" in Microsoft Word, editors can make suggestions that the author can then accept or reject individually.
- Real-time Comments: A comment thread will be available for discussions, supporting mentions to notify specific collaborators.

1.4.2 Audience

The findings and deliverables of this project will primarily benefit the following groups:

- Web Novel Authors: As the core users, they will directly benefit from a more professional and intelligent creative environment, allowing them to focus more on the art of storytelling.
- Content Editors and Planners: They can leverage the system's collaborative and analytical features to guide authors more effectively, assess manuscript quality, and gain a high-level understanding of a work's structure and potential.

1.5 Proposed Structure of the Report

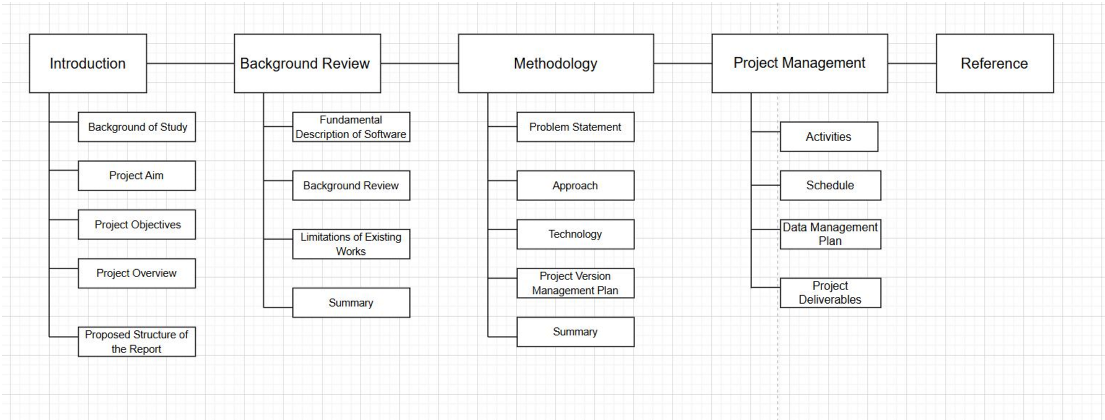

Figure 1-1. Project structure flowchart

Chapter 2. Background Review

2.1 Fundamental Description of Software

The ProsePal is a writing assistant software. It is a category of applications designed to help users improve their writing efficiency and quality. Core functionalities typically include text editing, formatting, and spell/grammar checking. With technological advancements, modern writing tools have evolved beyond basic word processing to integrate advanced features such as project management, research organization, distraction-free writing modes, and online collaboration to meet the specific needs of various domains [4]

2.2 Background Review

2.2.1 Comparative Analysis of Software Features

This section provides a comparative analysis of existing writing software relevant to this project. These tools can be categorized into three types: general document tools, professional writing software, and existing web novel writing tools. The analysis will focus on their features regarding document management, collaboration capabilities, and intelligent analysis.

Table 2-1. Format of Table

Authoring Tools	**Document Management (Version Control, Outline)**	**Collaboration Features**	**Intelligent Analysis (Character Mapping, Plot Curve)**	**Proofreading check**
Microsoft Word[6]	Yes	Yes	No	Yes
Google docs[7]	Yes	Yes	No	Yes
Scrivener[8]	Yes	Yes	No	No
Ulysses[9]	Yes	Yes	No	No
Writer's assistant	Yes	Yes	No	Yes
Orange melon writing	Yes	No	No	Yes
Our Proposed System	Yes	Yes	Yes	Yes

2.3 Limitations of Existing Works

The analysis above reveals a clear market gap in existing tools:

- General Document Tools lack professional features designed for long-form writing, such as complex chapter management and character tracking, leading to a fragmented creative process.
- Professional Writing Software, while excellent for structural management, generally has weak or non-existent collaboration features, failing to meet the close communication needs between authors and editors.
- Existing Web Novel Tools often have basic functionalities and weak analytical capabilities, failing to fully leverage data analysis techniques to assist creative decision-making.

In summary, the current market lacks an integrated solution that seamlessly combines a fluid writing experience, deep narrative intelligence, and efficient online collaboration.

2.4 Summary

This chapter introduced the fundamental concept of writing assistant software and, through comparative analysis, identified the shortcomings of existing tools in serving web novel authors. These limitations highlight the necessity of developing a more comprehensive and intelligent integrated writing platform, which is the core goal of this project.

Chapter 3. Methodology

This chapter details the methodology adopted to solve the aforementioned problems, including a clear problem statement, software development approach, technology selection, and version management plan.

3.1 Problem Statement

Web novel authors face three core problems in their creative process:

- Inefficiency due to Tool Fragmentation: Authors must frequently switch between software for writing, research, and outlining, which disrupts their creative flow and increases cognitive load [5].
- Lack of Macro-level Management for Complex Narratives: Long-form novels involve numerous characters and plotlines that are difficult to manage and track with traditional tools, leading to potential logical inconsistencies.
- Subjective and Inefficient Manuscript Revision Process: Collaboration between authors and editors relies on offline, unstructured communication that lacks data support, resulting in long and inefficient revision cycles.

3.2 Approach

3.2.1 Software Development Method

This project will adopt Agile Development principles, closely following a Scrum framework. Each sprint will begin with a planning meeting to select tasks from a product backlog and will conclude with a sprint review to demonstrate a working, incremental version of the software. This iterative approach is ideal for this project as it allows for flexibility and rapid prototyping.

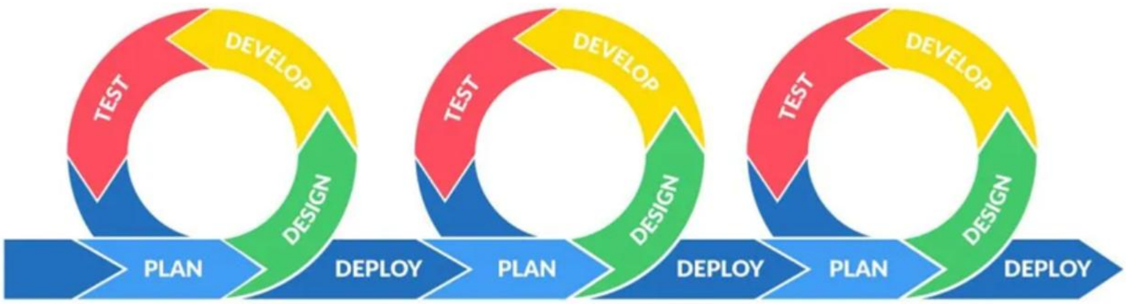

Figure 3-1. Image of Agile

3.2.2 Requirements Gathering

A multi-faceted approach will be used for requirements elicitation:

- Indirect Interviews: Analyzing interview records with web novel editors to extract core pain points and feature needs.
- Competitive Analysis: Systematically analyzing existing writing software to establish functional baselines and identify opportunities for innovation.
- User Personas: Creating typical user personas of web novel authors to ensure design decisions remain user-centric.

3.2.3 Testing and Evaluation Techniques

- Unit Testing: Using frameworks like Jest to test individual components in isolation.
- Integration Testing: Verifying that different modules interact correctly.
- User Acceptance Testing: Inviting authors and editors to test the software in later stages to gather qualitative feedback for final optimization.

3.3 Technology

Table 3-1. Table of software and hardware

Category	Tools	Purpose
Frontend[10]	Vue.js+TypeScript	Building a responsive and interactive user interface.
Backend	Node.js[11],Python[12]	Handling business logic, API requests, and user authentication.
Database	MongoDB[13],MySQL	Storing document content, user data, and analysis results.
Search Engines	Elasticsearch	Fast full-text search and keyword positioning
AI&NPL service	DeepSeek，spaCy[14]，ECharts，SnowNLP	DeepSeek/spaCy+ECharts timeline visualization, emotion curve using SnowNLP+ECharts,
Proofreading	Grammarly API[15]	Grammar and spelling error checks
Real-time	WebSocket[16], Socke t.IO [17]	Real-time data push and communication

Hardware	Developer PC,Cloud Sever	Develop and test locally; Deploy to backend and production environments
Testing Tools	Jest[18],Postman[19]	Unit test, API test

3.4 Project Version Management Plan

The development process will use Git for version control, with GitHub as the remote code repository. The development will be divided into four main versions.

Table 3-2. Table of version management plan

Versions	Details
Version 1	Implementation of the core writing MVP
Version 2	Integration of basic intelligent analysis features.
Version 3	Introduction of collaboration and search features.
Version 4	Implementation of advanced analysis features and product optimization.

The full project source code will be available at: https://github.com/jxlin0524/Web-application-project

3.5 Summary

This chapter has defined the core problems the project aims to solve. By combining established requirements engineering practices, a comprehensive testing strategy, and a modern technology stack, a solid foundation has been laid for the project's successful implementation.

Chapter 4. Project Management

4.1 Activities

Table 4-1. Table of activities

Activities	Details
Preliminary Research	Determine the research topic, collect and analyze relevant academic literature and market competitors.
Prepare Project Proposal	Complete the Project Proposal document.
Requirement Analysis	Research requirement, find suitable methods to solve the problem
System Design	Design the detailed functional specifications, database schema, and system architecture.
Development & Implementation	Code the frontend, backend, and NLP services according to the version plan.
Testing & Evaluation	Perform unit, integration, and user acceptance testing for all software modules.
Thesis Writing	Write the final thesis
Thesis Modify	Make thesis reversions
Presentation Preparation	Prepare the final project presentation slides and defend the project.

4.2 Schedule

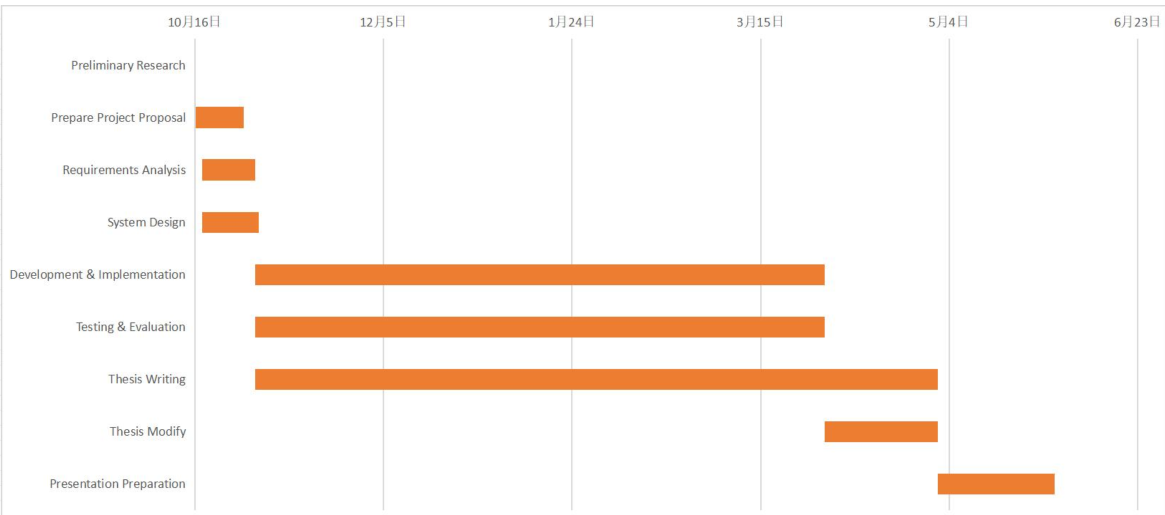

Figure 4-2. Project Schedule

4.3 Data Management Plan

Zotero is used to manage the preliminary literature of the project, GitHub is used to manage the code and Baidu drive is used to manage the report.

Table 4-2. Caption should be here

Management Category	Tool
Literature Management	Zotero
Code Management	GitHub
Project Reports & Logs	Baidu drive

4.4 Project Deliverables

This section lists what needs to be delivered throughout the project.

- The project proposal
- Weekly report
- Progress Report
- Final Project Report
- Project codes
- Project presentation slides
- Project presentation

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